

O2C MAD LABS - Research Mesh Report

Autonomous Research Cycle: How can Physics-Informed Neural Networks (PINNs) be used to

Key Metrics

- Components: 32
- Tests Passing: 260
- Papers Ingested: 154
- Papers Distilled: 20
- Graph Nodes: 283
- Findings: 1
- Cycle Errors: 0
- Duration: 16.9s

Research Question

The autonomous research cycle was tasked with investigating: "How can Physics-Informed Neural Networks (PINNs) be used to trade or map volatility?"

This was deliberately chosen as a stress test. The two domains have no obvious surface-level connection, making it an ideal test of the research mesh's ability to discover latent cross-domain relationships.

Pipeline Execution

Step	Status	Details
?	PASS	total_papers=154, openalex=133, arxiv=20
?	PASS	distilled=20, graph_nodes=283, graph_edges=20527

?	PASS	gaps_found=0, gaps=[]
?	PASS	task_id=task_24cf547d26a6, query=research in pinns volatility, findings=1, fi...
?	PASS	nodes_added=444, edges_added=20527, papers_synced=83, doctrine_synced=135, er...
?	PASS	

L1 ? Paper Ingestion

Papers were ingested from OpenAlex and arXiv across two domains: PINNs (Physics-Informed Neural Networks) and Volatility Trading.

- Total papers in database: 154
- OpenAlex papers: 133
- arXiv papers: 20

Top Papers by Citation Count

Title	Source	Citations	Status
MizAR 60 for Mizar 50	openalex	76066	distilled
ImageNet classification with deep convolutional n	openalex	75705	distilled
Gradient-based learning applied to document rec	openalex	57987	pending
Global cancer statistics	openalex	55035	pending
	openalex	50260	distilled
Emergence of Scaling in Random Networks	openalex	36303	distilled
A new criterion for assessing discriminant validit	openalex	33161	pending
Array programming with NumPy	openalex	21925	distilled
Global Cancer Statistics, 2002	openalex	18384	pending
Physics-informed neural networks: A deep learnir	openalex	16399	distilled

L2 ? Distillation + Knowledge Graph

20 papers were distilled into structured vault notes using the CAUSE/METHOD/RESULT/LIMITATIONS/APPLICATION/LINKS format.

- Graph nodes: 283
- Graph edges: 20527

Graph Composition

Node Type	Count
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paper	143
doctrine	135
concept	29

L3 ? Research Agent Findings

Research agent task: research in pinns volatility
Task ID: task_24cf547d26a6
Findings: 1

Cross-Domain Findings

Fractional Brownian Motions, Fractional Noises and Applicati (confidence: 0.76)
Source: openalex
Relevance: Cross-domain bridge: fBm in stochastic PDEs (PINNs) and volatility modeling (finance)

L4 ? Vault Sync + Knowledge Graph

Vault sync completed: 444 nodes and 20527 edges added to the knowledge graph.

- Papers synced: 83
- Doctrine notes synced: 135

Telemetry + Safety Status

All safety caps and budgets after the autonomous cycle:

Safety Rule	Limit	Used	Remaining
LLM Budget	\$2.00	\$0.00	\$2.00
Vault Writes	200	0	200
Agent Slots	3	0	3

Errors

Clean Run
No errors were encountered during the autonomous cycle.

Component Inventory (32 Components)

Layer	File	Component	Agent	Tests
L1	openalex_client.py	OpenAlex API	PM	15 tests
L1	arxiv_client.py	arXiv API	PM2	6 tests
L1	s2_client.py	Semantic Scholar	PM	10 tests
L1	sources.py	Source Registry	CC	-
L1	models.py	Paper Schema	CC	-
L1	scheduler.py	Scheduler	RL	-
L1	cache.py	Cache + Dedup	PM	6 tests
L1	rate_limit.py	Rate Limiter	PM2	5 tests
L2	distiller.py	Distiller	CC	-
L2	concepts.py	Concept Extractor	PM	-
L2	citation_graph.py	Citation Graph	PM2	-
L2	vault_writer.py	Vault Writer	CC	-
L2	graph_store.py	Graph Store	CC	-
L2	llm_distill.py	LLM Distiller	AS	-
L2	doctrine.py	Doctrine Extractor	AS	-
L2	contradictions.py	Contradiction Detector	RL	-
L3	gap_detector.py	Gap Detector	AS	-
L3	task_gen.py	Task Generator	PM	-
L3	research_agent.py	Research Agent	CC	-
L3	evaluator.py	Finding Evaluator	PM2	-
L3	router.py	Task Router	PM2	-
L3	lifecycle.py	Agent Lifecycle	AS	-
L3	queue.py	Task Queue	CC	-
L3	srta_adapter.py	SRRA Adapter	CC	-
L4	research_api.py	API (18 endpoints)	CC	35 tests
L4	vault_sync.py	Vault Sync	PM2	-
L4	telemetry.py	Telemetry + Audit	AS	-
L4	researchStore.ts	Frontend Store	PM2	-
L4	research/page.tsx	Research Hub	PM2	-
L4	research/graph/	Knowledge Graph UI	PM2	-
L4	research/doctrine/	Doctrine Library UI	PM2	-
L4	research/agents/	Research Agents UI	PM2	-

Test Summary

Complete test coverage across all layers:

Test Suite	Count	Coverage
L1 Ingestion (unit)	46	All source clients, cache, rate limiter
Safety Regression	41	6 hard rules, daily caps, status transitions
L2 Integration	24	Distiller, vault writer, graph, concepts, citations, doctrine
L3 Integration	26	Gap detector, task gen, queue, evaluator, router, lifecycle
Cross-layer	88	End-to-end pipelines, safety boundaries
L4 API	35	All 18 endpoints, response validation
TOTAL	260	Full stack coverage

O2C MAD LABS Sovereign Research Mesh - Build Report

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